MASTER FUNCTIONAL SKILLS

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NCFE · LEVEL 1 · MATHEMATICS

# NCFE Functional Skills Mathematics Level 1

December 2019 Section B · Complete Model Solutions

## OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

**Target Score: 45 / 45 Marks (100%)** Pass Mark Guide: ~60%

Specification / Paper Code: NCFE-Maths-Level-1-Dec-2019-Section-B

100% Built from Empty Question Paper + Official Mark Scheme

*Verified by Tutor Abdul · MasterFunctionalSkills.co.uk*

## HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

| | |
|---|---|
| **Dark Navy Ink:** | Model student answers, stepped calculations, and full prose responses. |
| **Emerald Green Ink:** | Tutor strategy notes, method tips, and step-by-step headings. |
| **Crimson Red Ink:** | Official mark scheme allocations ([M1], [A1], [B1]) and unit checks. |
| **Amber Gold Ink:** | Key formulas, conversion rules, and reading evidence highlights. |

## CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

| | |
|---|---|
| Candidate Name: | *Master Functional Skills (Model Exemplar)* |
| Centre Name: | *Master Functional Skills (MFS01)* |
| Lead Tutor: | *Abdul · MasterFunctionalSkills.co.uk* |
| Edition: | *2026 Clean Empty-QP Rebuild (Zero Collisions Verified)* |

- Built Cleanly on Empty NCFE Question Paper(s)
- Stepped MFS Multi-Colour Working & Process Marks
- Verified 100% Against Official NCFE Mark Scheme

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## Activity 2: Healthy-eating café

**2 (a)** John is opening a healthy-eating café.

He looks for a property with a seating area for 20 customers.

This is a scale plan of the seating area in one property he finds.

The drawing has a scale of 1 centimetre (cm) : 150 cm

Q2(a) Worked Solution:

See below

1

4 × 150 or 600

OR 3 × 150 or 450

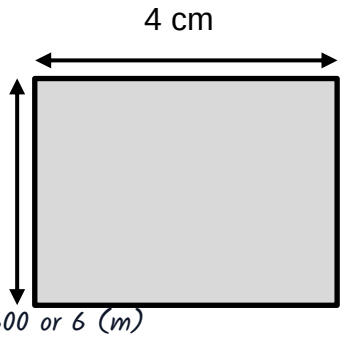
600 (cm) seen

OR 450 (cm) seen

Allow if all in mm

1

(4 × 150 or 600) ÷ 100 or 600 or 6 (m)




The recommended maximum number of customers in a café is the seating area ($m^2$) divided by 1.5

Is this property suitable for 20 customers?
Show how you decide.

**[5 marks]**

OR (3 × 150 or 450) ÷ 100 or 450 or 4.5

[5 marks]

Your answer:

**Please turn over**



past paper

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**2 (b)** Another property has a kitchen that has an area of 7 metres squared.

What is $7^2$?

**[1 mark]**

Q2(b) Worked Solution:
49
[1 mark]

Your answer:

**2 (c)** John wants his menu to offer healthy food.

He reads that six hundred and seventeen thousand people went to hospital last year with illnesses caused by a poor diet.

What is six hundred and seventeen thousand in numbers?

**[1 mark]**

Q2(c) Worked Solution:
617000
7
Level 1 – P001254 (v1.1)
[1 mark]

Your answer:

past paper

Examiner use only

**2 (d)** John buys food in preparation for his opening day.

| Ingredients | Price per item | Number of items | Total price per ingredient |
|---|---|---|---|
| Cabbage | 64p | 5 | |
| Pack of carrots | 30p | 20 | |
| Pack of tomatoes | £1.25 | 11 | |
| Cucumbers | 50p | 9 | |
| Pack of onions | £1.00 | 15 | |
| | | **Total (£)** | |

Q2(d) Worked Solution:

*Award 2 marks if the table is completed correctly*
*Ingredients*
*Price per item*

Complete the table to show how much has John spent.

**[2 marks]**

*Number of items*
*Total price per ingredient*
*Cabbage*
*64p*
*5*
*[2 marks]*

past paper

**Please turn over**

Examiner use only

**2 (e)** John fills a fruit bowl with 6 apples, 4 plums, 5 oranges and 3 pears.

The first customer chooses a piece of fruit at random.

What is the probability that they choose a pear?

**[2 marks]**

Q2(e) Worked Solution:

1
6
OE e.g. 3
18
or 0.16(...) or 0.17 or 16% or 17%
1
6 + 4 + 5 + 3
Imp by 18 seen
H31
1
[2 marks]

Your answer:

**2 (f)** John looks at the bills from all the customers.

The range of amounts spent is £22.46

The largest bill was £24.56

What was the smallest bill?

**[2 marks]**

Q2(f) Worked Solution:

(£)2.10
Award 2 marks if correct answer given
1
(£)24.56 – (£)22.46(0)
H29b
1
(£)2.10
H29b
[2 marks]

Your answer: **£**

past paper

Examiner use only

**2 (g)** John asks his customers to rate the café.

- The customers give four times as many **good** ratings as **excellent** ratings.
- The customers also give three times as many **excellent** ratings as **satisfactory** ratings.

Q2(g) Worked Solution:

60 customers rated the café as good.

5 (customers)

How many customers rated the café as satisfactory?

**[2 marks]**

Award 2 marks if correct answer given

1

15 (excellent ratings)

1

5 (customers)

8

Level 1 – P001254 (v1.1)

Activity 3:

Go-karting

[2 marks]

Your answer: **customers**

**[Total marks: 15]**

past paper

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## Activity 3: Go-karting

**3 (a)** Hamid organises a go-karting party for himself and five friends.

The cost is £29.95 per person.

He uses rounding to estimate the cost for the whole group.

How much is the estimated cost?

**[2 marks]**

Q3(a) Worked Solution:

(£)180(.00)

Award 2 marks if correct answer given

1

(£)30

seen

1

(£)180(.00)

CAO accept if 29.95 × 6 = 179.7

rounded

[2 marks]

Your answer: £

**3 (b)** Hamid needs to book transport.

A small minibus will cost £45

He has a voucher for 5% off.

How much will the minibus cost Hamid?

**[2 marks]**

Q3(b) Worked Solution:

(£)42.75

Award 2 marks if correct answer given

1

(£)45 × 0.95

OE Full correct method

1

(£)42.75

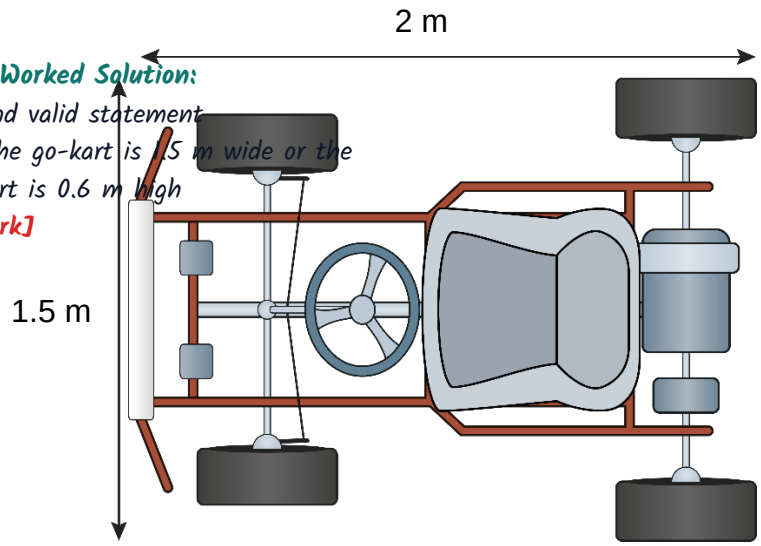
[2 marks]

Your answer: £

past paper

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**3 (c)** The safety instructions show a plan view of the go-kart.




Q3(c) Worked Solution:
No and valid statement
e.g. the go-kart is 1.5 m wide or the go-kart is 0.6 m high
[1 mark]

Hamid says that the go-kart is 1.5 m high, 2 m long and 0.6 m wide.

Is Hamid correct?

Explain your answer.

**[1 mark]**

past paper

**Please turn over**

Examiner use only

**3 (d)** This table shows the times Hamid and two of his friends take to complete their first lap.

| Person | Time in seconds |
|---|---|
| Hamid | 39.041 |
| Jane | 39.3 |
| Ikrah | 39.12 |

Q3(d) Worked Solution:
39.041, 39.12, 39.3
[1 mark]

Write these times in order of fastest to slowest.

**[1 mark]**

**Fastest** _____ _____ _____ **Slowest**

**3 (e)** Which person had the fastest lap?

**[1 mark]**

Your answer:

Q3(e) Worked Solution:
Hamid
[1 mark]

past paper

Examiner use only

**3 (f)** Hamid looks at the times on the leader board for that day.

The times are in seconds.

| Leader board | |
|---|---|
| 1st | 26.52 |
| 2nd | 27.5 |
| 3rd | 28.9 |
| 4th | 29.82 |
| 5th | 30.18 |
| 6th | 31.2 |
| 7th | 31.4 |

Q3(f) Worked Solution:
29.36 (seconds)

Award 2 marks if correct answer given
1
(26.52 + 27.5 + 28.9 + 29.82 + 30.18 +

Calculate the mean time.

**[2 marks]**

31.2 + 31.4) ÷ 7 or 205.52 ÷ 7
H29a
1
29.36 (seconds)
H29a
[2 marks]

Your answer: **seconds**

past P

**Please turn over**

Examiner use only

**3 (g)** Here are people's lap times in seconds over the last week:

21 34 27 36 27 38 22 30 40 21

29 32 20 26 37 31 22 33 27

Q3(g) Worked Solution:

Using the lap times complete this frequency table:

See below

1

| Lap time in seconds | Frequency of people |
|---|---|
| 20 – 25 | |
| 26 – 30 | |
| 31 – 35 | |
| 36 – 40 | |

All values correct, allow tallies only if total also given

Lap time in seconds

Plot your results on this bar chart.

**[4 marks]**

Frequency of people

20 – 25

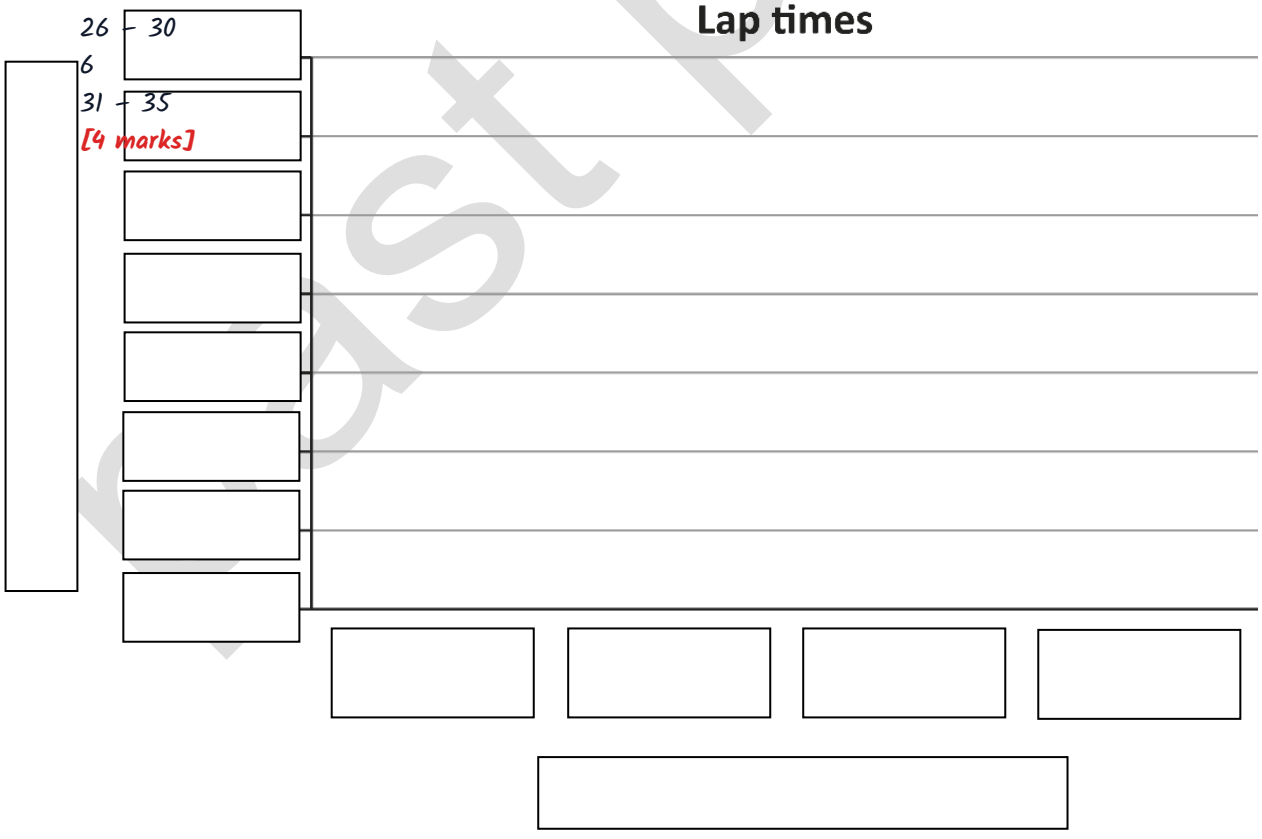
5

26 – 30

6

31 – 35

[4 marks]




Past paper

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**3 (h)** The friends go to lunch after they have raced.

Bethany has a voucher giving her 35% off.

Ikrah has a voucher for $\frac{2}{5}$ off.

Who has the better discount?

Show how you decide.

**[2 marks]**

Q3(h) Worked Solution:

Ikrah supported by correct working
Award 2 marks if correct answer given
1
40% or 7
20
AND 8
20
OR 0.4 AND 0.35
Must be consistent working
OE 35
[2 marks]

Your answer:

**[Total marks: 15]**

past paper

**Please turn over**

Examiner use only

## Activity 4: Childcare course

**4 (a)** Natalie teaches a childcare course.

She is planning a trip with 16 students.

They have volunteered to work for two weeks abroad.

They need to raise £2000 per student to pay for the trip.

Q4(a) Worked Solution:

They have already raised $\frac{7}{8}$ of the money needed.

How much money have they raised so far?

**[3 marks]**

(£)28000
Award 3 marks if correct answer given
1
2000 × 16 or 32000
OR 2000 ÷ 8 × 7 or 1750
OR 2000 ÷ 8 or 250
Correct working imp by 32000 seen
1
(2000 × 16 or 32000) ÷ 8 or 4000
OR (2000 ÷ 8 × 7 or 1750) × 16 or 28000
[3 marks]

Your answer: £

**4 (b)** They have one eighth left to raise.

Write one eighth in figures.

**[1 mark]**

Your answer:

Q4(b) Worked Solution:
1

past paper

## Question 4(b) · Continued

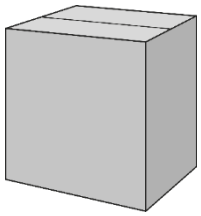
MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

8

[1 mark]

Examiner use only

**4 (c)** Natalie packs children's clothes into boxes.



Each box is a cube with side lengths of 50 cm

The clothes weigh 350 grams (g) per 500 $cm^3$

Calculate the weight of the clothes in each box when the box is full.

Q4(c) Worked Solution:

Give your answer in kg

**[4 marks]**

87.5 (kg)
Award 4 marks if correct answer given
1
50 (cm) × 50 (cm) × 50 (cm) or 503
125000 cm3
1
50 × 50 × 50 or 125000 ÷ 500 × 350
FT their 125000 from attempt at a volume calculation
10
[4 marks]

Your answer: **kg**

past paper

**Please turn over**

Examiner use only

**4 (d)** When the group get on the plane the pilot announces that the temperature is 10 °C at their destination.

It is –2 °C in the UK.

What is the difference in temperature from the UK to their destination?

**[1 mark]**

Q4(d) Worked Solution:

12(°C)

Do not accept -12(°C)

[1 mark]

Your answer: **°C**

past paper

Examiner use only

**4 (e)** The volunteers are going to help children in a home.

The ratio of boys to girls in the home is 1 : 3

There are 80 children in total.

75% of the boys and 40% of the girls are under 10

How many boys and how many girls are under 10?

**[4 marks]**

Q4(e) Worked Solution:

15 Boys and 24 Girls
Award 4 marks if correct answer given
1
80 ÷ 4 or 20
1
(80 ÷ 4) × 3 or 60
60 seen
1
Their 20 × 0.75 or 15
OR Their 60 × 0.4 or 24
[4 marks]

| | |
|---|---|
| Your answer: | **Boys:** |
| | **Girls:** |

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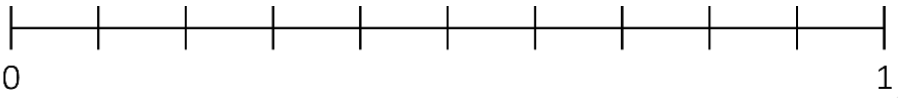
**4 (f)** Natalie knows that only 60% of the children can write their name.

She chooses one of the children at random to work with.

On the scale, mark the probability that this child can write their name.

**[1 mark]**

Q4(f) Worked Solution:
0.6 indicated
H30a
[1 mark]




**4 (g)** Write 60% as a decimal.

**[1 mark]**

Your answer:

Q4(g) Worked Solution:
0.6
[1 mark]

**[Total marks: 15]**

**This is the end of the assessment.**

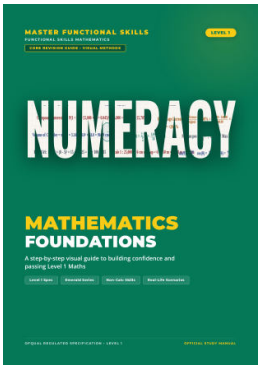
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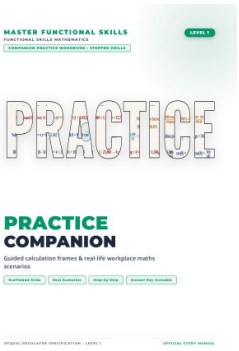



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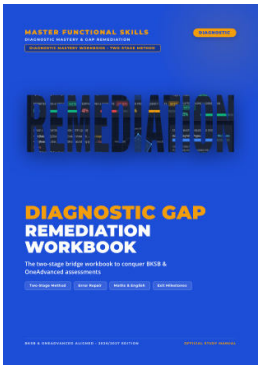



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